Individual Assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(here)
```

```
here() starts at /Users/maisieprestridge/Desktop/github/week-05_spring-
2026_aquatic-inverts
```

```
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(snakecase)

# wrap axis labels
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
Rows: 50 Columns: 15
-- Column specification --------------------------------------------------------
```

```
Delimiter: ","
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
```

```
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and taxon.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

x-axis: site_full y-axis: log_ave_lit geometry to represent average abundance/L: geom_jitter() geometry to represent medians: geom_point()

b. Wrangle your data

```
# new object created from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "ostracod") |>
  # log + 1 to transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods)
```

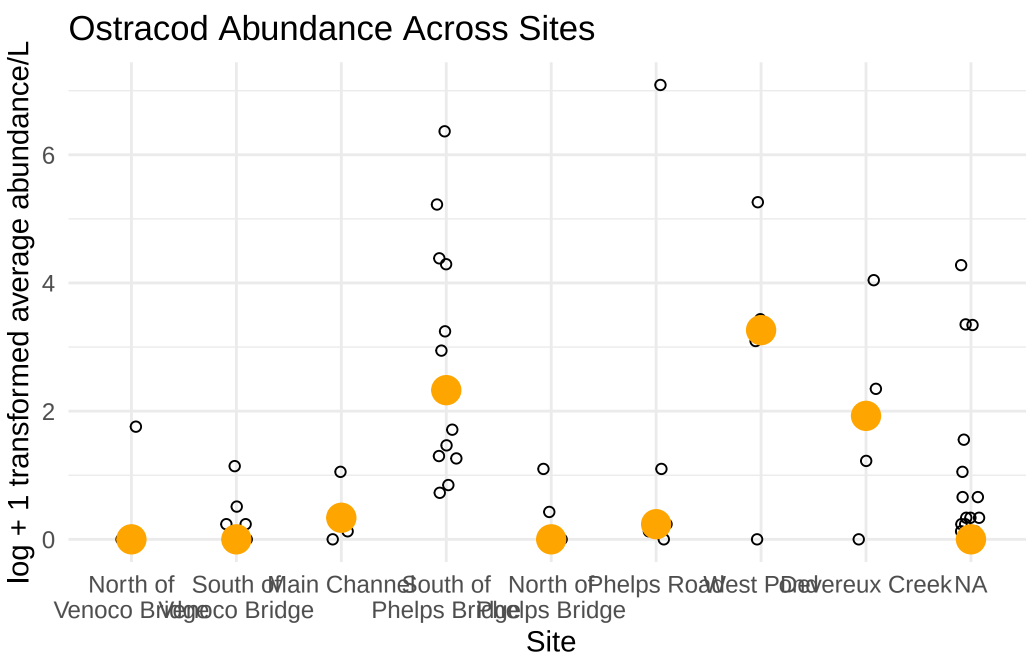
```
# A tibble: 1 x 24
  date_site       date                    site   taxon   ave_lit med_ph med_sal med_do
```

```
  <chr>           <dttm>              <chr> <chr>    <dbl>  <dbl>   <dbl>  <dbl>
1 2021-11-17_CUL1 2021-11-17 00:00:00 CUL1  ostra~       0   7.77    8.83   6.52
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```r
# base layer
ostracodgraph <- ggplot(data = ostracods,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = log_ave_lit)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "orange",
               size = 1) +
  # labelling x-axis and y-axis
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod Abundance Across Sites") +
  # change theme from default
theme_minimal()

# displaying object
ostracodgraph
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_segment()`).
```

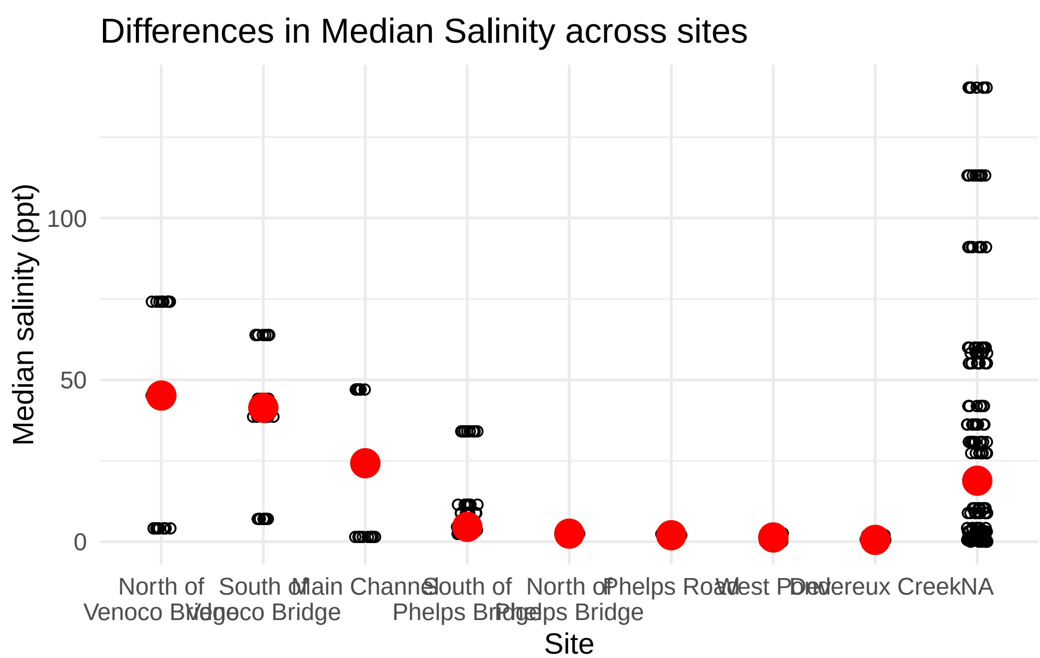


d. Create a multipanel figure

```
# salinity
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # labellingg x-axis and y-axis
  labs(x = "Site",
       y = "Median salinity (ppt)",
```

```
      title = "Differences in Median Salinity across sites") +
  # different theme
theme_minimal()
# displaying object
salinity
```

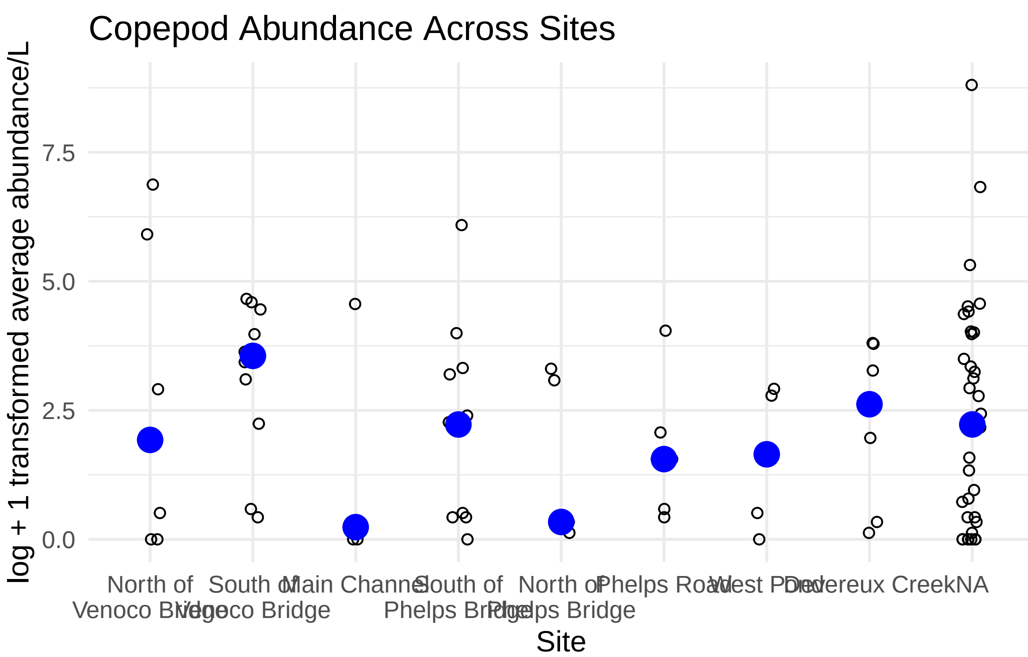
```
Warning: Removed 396 rows containing non-finite outside the scale range
(`stat_summary()`).

Warning: Removed 396 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_segment()`).
```




```
# cleaning copepods before graphing
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
8
```

```
[1] 8
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod Abundance Across Sites") +
  # different theme
theme_minimal()
# displaying object
copepods
```



e. Interpret your figure

2. Visualizing total abundance as a function of salinity